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Characterization of High Performance Third Generation Current Conveyor using CMOS Technology

Megha M. Patel[1]*, Nilesh D. Patel[2]
[1]Department of VLSI System Design, L. C. Institute of Technology,
Bhandu, Gujarat, India
[2]Department of VLSI System Design, Institute of Technology,
Nirma University, Ahmedabad, India

Abstract
In this paper, Conventional Third generation(CCIII) and Dual-Output Cascoded Third Generation Current Conveyor(DOCCIII) is presented. Here, both are simulated in ELDO SPICE Mentor Graphics using 90 nm CMOS technology at 0.8 V supply voltage. High performance cascoded current mirrors provide high output impedance. Third generation current conveyer has the advantages of wide current and voltage bandwidths, controlled intrinsic resistances at terminals X,Y and Z. By the use of high performance dual outputcascoded current mirrors provides high output impedance 21.25 MΩ and13.47MΩ instesd of inconventional CCIII output impedance 4.79 KΩ and voltage bandwidth is 704 MHz and current bandwidths are 300 MHz and greater than 1 GHz. In cascoded dual output third generation current conveyor, power dissipation is 6.94 µWand gives excellent input/output current gain.

Index Terms:*CMOS Circuits,CCIII, DOCCIII, ELDO SPICE Mentor Graphics, power dissipation*

****Author For Correspondence*** E-mail: meggi2742@gmail.com

INTRODUCTION
Fabre proposed third generation current conveyor in 1995. It can be considered as a unity gain current controlled current source. The main features of the CCIII are high accuracy, high output impedance,wide frequency response and High swing. High output impedance at terminal Z of the CCIII is required to enable easy cascadability and easy structure without need for additional active elements in applications[1–3]. High performance current mirrors are required in the structure of the CCIII to provide good dynamic swing and high output resistance which enables cascadability[4–7]. In this paper, we propose a novel implementation of dual output CCIII based on an improved active-feedback cascode current mirrors. ThecascodedCCIII is compared with the conventional CCIII. It provides high output resistance at port Z and high dynamic swing.

CIRCUIT DESCRIPTION
The Port relations of an ideal dual-output CCIII,which is shown in Figure 1,can be given by,

$$\begin{pmatrix} \mathbf{Iy} \\ \mathbf{Vx} \\ \mathbf{Iz+} \\ \mathbf{Iz-} \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{Vy} \\ \mathbf{Ix} \\ \mathbf{Vz+} \\ \mathbf{Vz-} \end{pmatrix} \qquad \mathbf{(1)}$$

By above equation, an ideal dual-output CCIII has a (+1) voltage gain between terminals X and Y,a (-1) current gain between terminals X and Y, a (+1) current gain between terminals X and +Z, and a (-1) current gain between terminals X and –Z.

Where, the positive sign for positive current conveyor and negative sign for negative current conveyor[8][10].

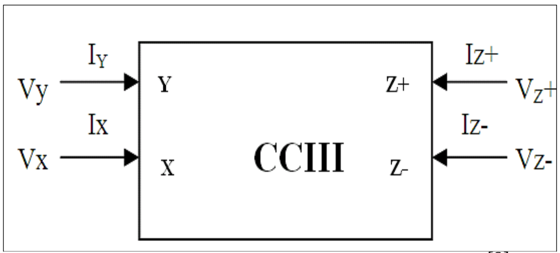

Fig. 1: *Electrical Symbol of the CCIII[8].*

The conventional third generation current conveyor is shown in Figure 2. It is based on basic current mirrors trans linear loop (M1,M2), (M5,M6), (M9,M10) and (M13,M14) which improve the bandwidths. By this output stage transistors M17- M18, current in ports X and Y is transferred to the ports Z. Here,the transistors between terminals X and Y works as voltage follower and the transistors between terminals X and Z works as current follower.A major advantage of this Third Generation Current Conveyor is its simple structure.The output resistance of this current conveyor is:

$$\mathbf{R_{oz}=(r_{ds17})//(r_{ds18})} \qquad \mathbf{(2)}$$

Where,r_{dsi} suggests the output resistance of the i'th transistor respectively. The disadvantage of using conventional CCIII is it's finite output resistance.

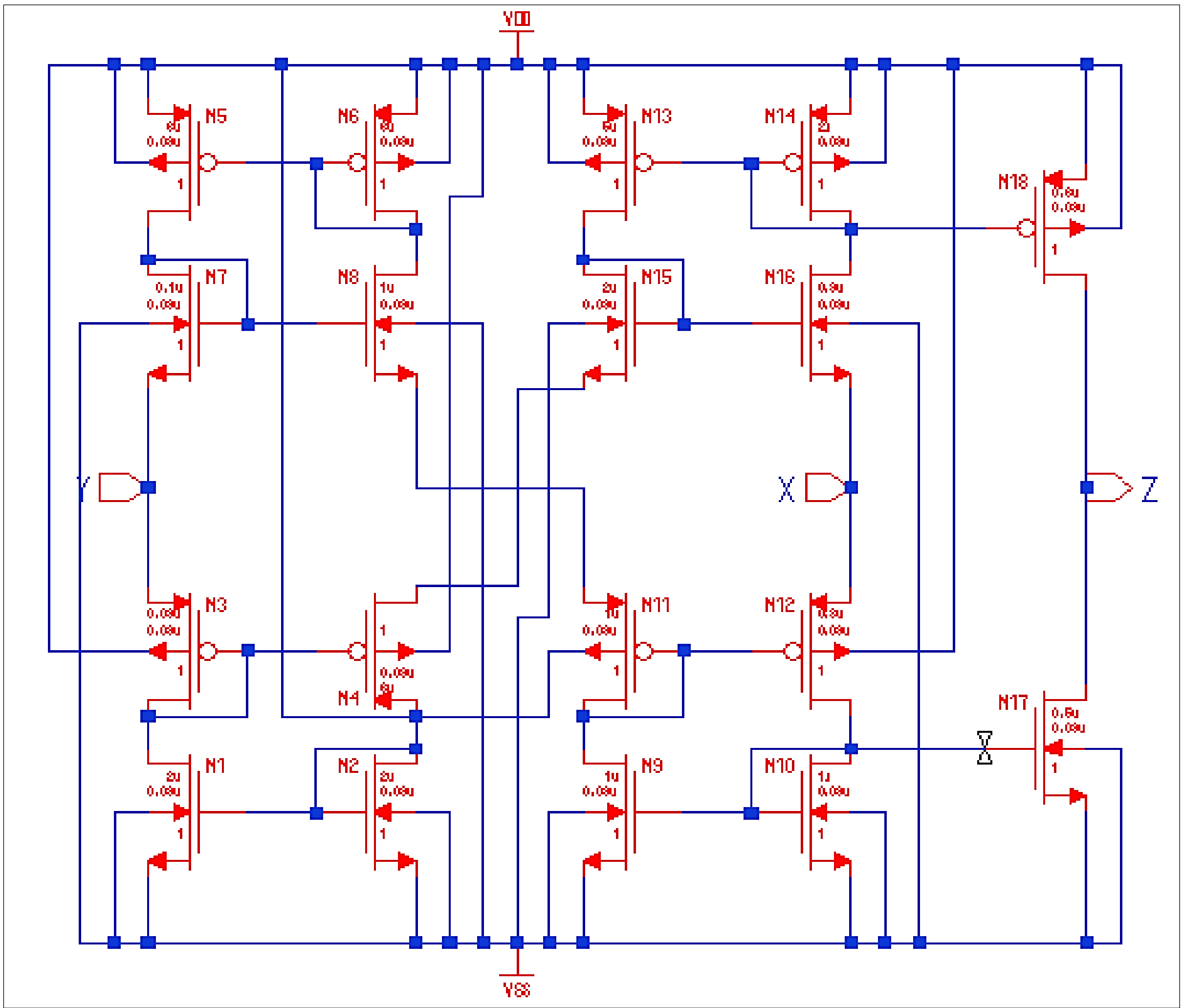

***Fig. 2:** Conventional Third Generation Current Conveyor(CCIII)*[8].

The dual output third generation cascoded current conveyor is shown in Figure 3. It is based on cascoded current mirrors consist of transistors (M21, M23, M25, M28), (M15, M20, M29, M30), (M5, M6, M0, M4), (M7, M8, M9, M10) are used to transfer the current between terminals X and Y.The current mirror used in the output stages consist of transistors (M11, M12) and (M31, M32) for Z+ and transistors (M1, M2) and (M17, M18) for Z-. By using dual output cascode current mirrors third generation current conveyer, current mirrors between ports X-Y, X-Z+, and Y-Z- as shown in Figure 3. The Z+ output resistance of the cascode CCIII can be found as:

$$\mathbf{R_{oz+}=(r_{ds12}\, r_{ds11}\, g_{m12})//(r_{ds31}\, r_{ds32}\, g_{m31})} \qquad \mathbf{(3)}$$

Comparing (2) and (3) it can be realized that the output resistance of the cascode CCIII is much higher than conventional CCIII.Although the output resistance of the cascode CCIII is higher than the conventional one.

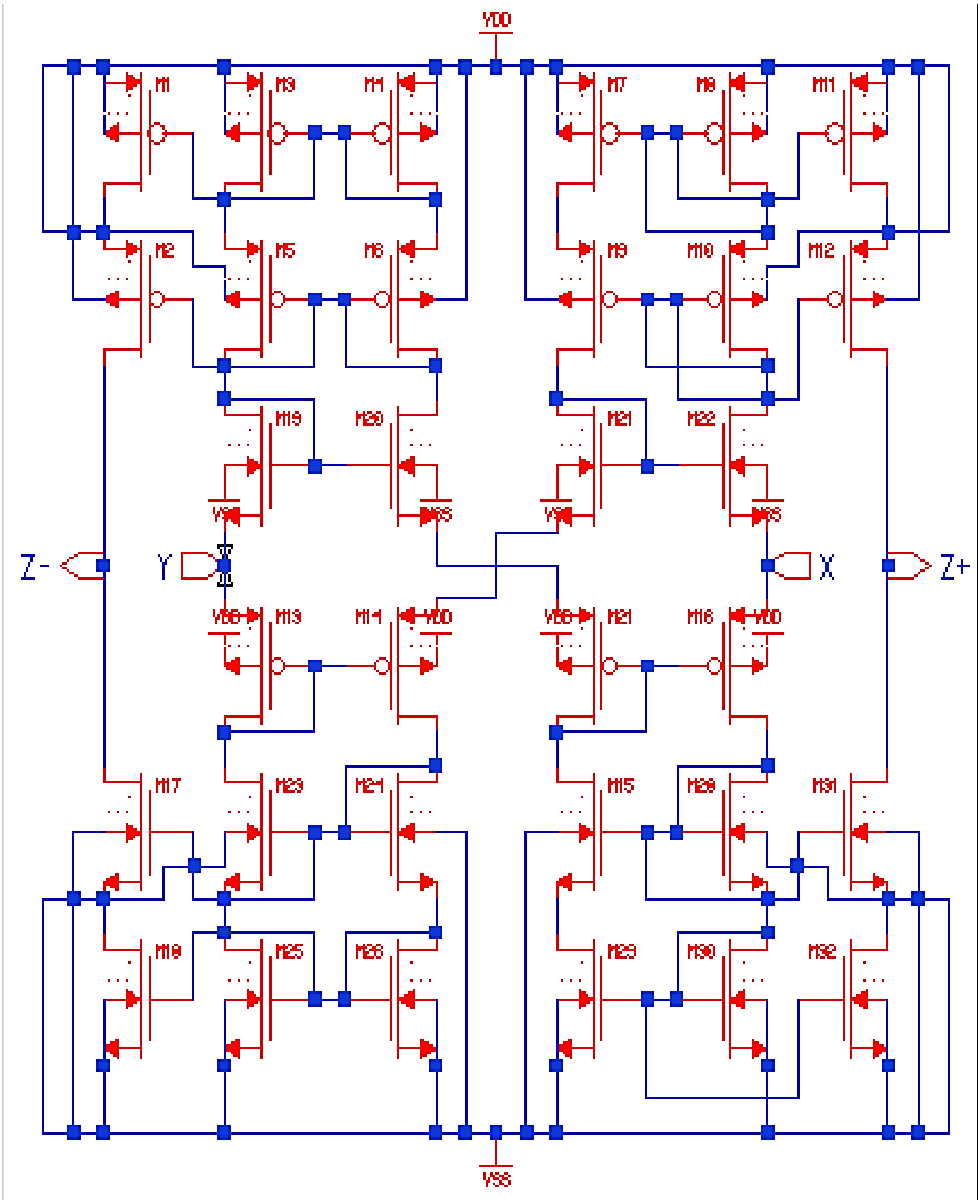

Fig. 3: *Dual Output Third Generation Current Conveyor using Cascode Current Mirrors(DOCCIII)[8].*

SIMULATION RESULTS AND COMPARISON

The performance of the conventional CCIII and cascoded CCIII shown in Figure2 and Figure3, is verified by ELDO SPICE Mentor Graphics tool using 90nm technology with supply voltage 0.8V.The main DC and AC characteristics of the CCIII, frequency responce of Vx/Vy and Iz+/Ix and Iz-/Ix.The AC transfer characteristics of voltaget gain and current gain of conventional CCIII is shown in Figure 4 and 5, respectively.

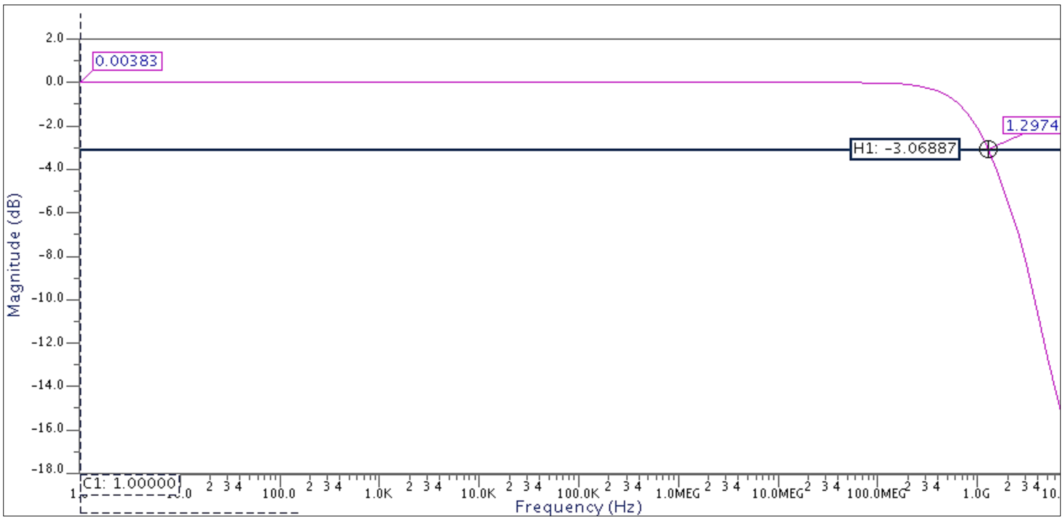

Fig. 4: *Frequency Response of the Vx/Vy.*

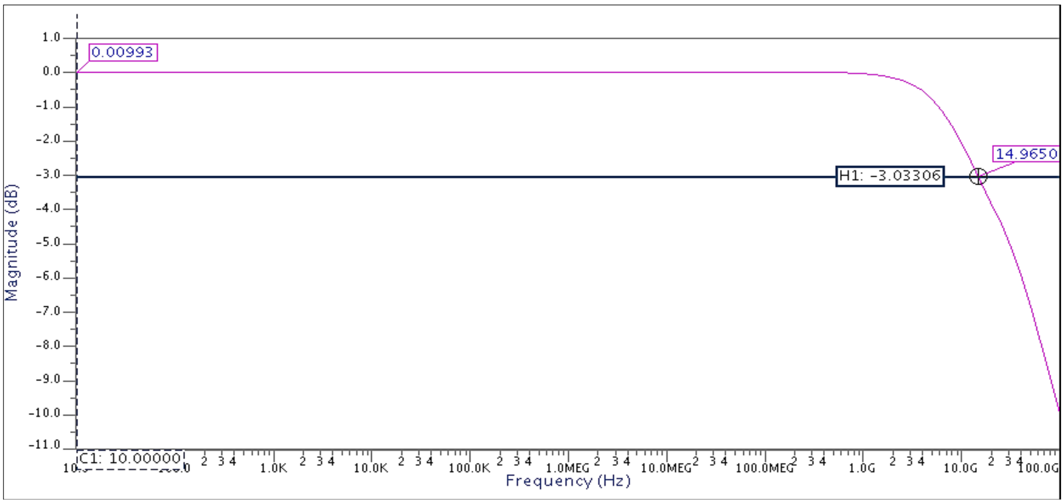

Fig. 5: *Frequency Response of the Iz/Ix.*

From the above Figure, it is shown that voltage bandwidth and cutternt bandwidth respectively 1.297 GHz and 14.96 GHz has been achieved.And X-port,Y-port and Z-port impedance is shown in Figure 6, 7 and 8 respectively.

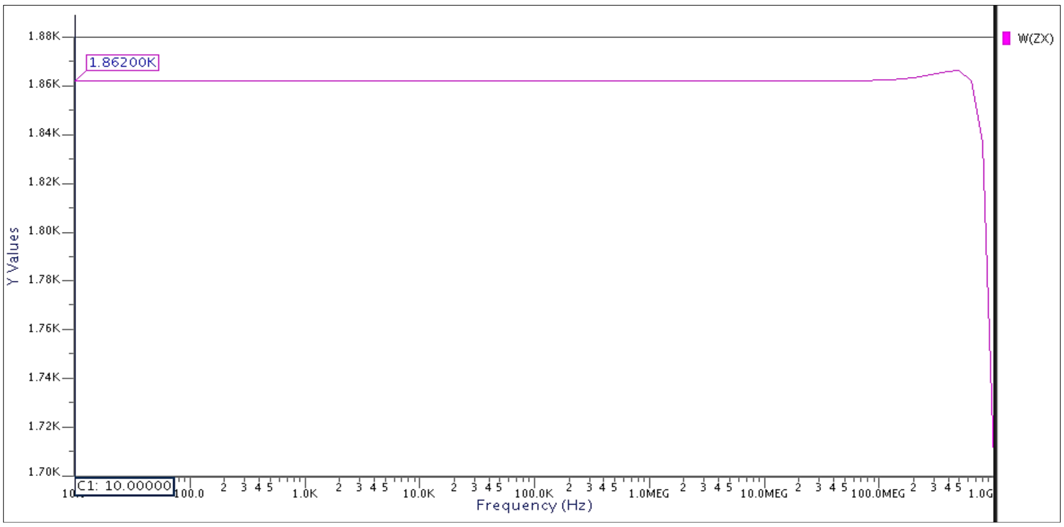

Fig. 6: *Frequency Response of the Zx.*

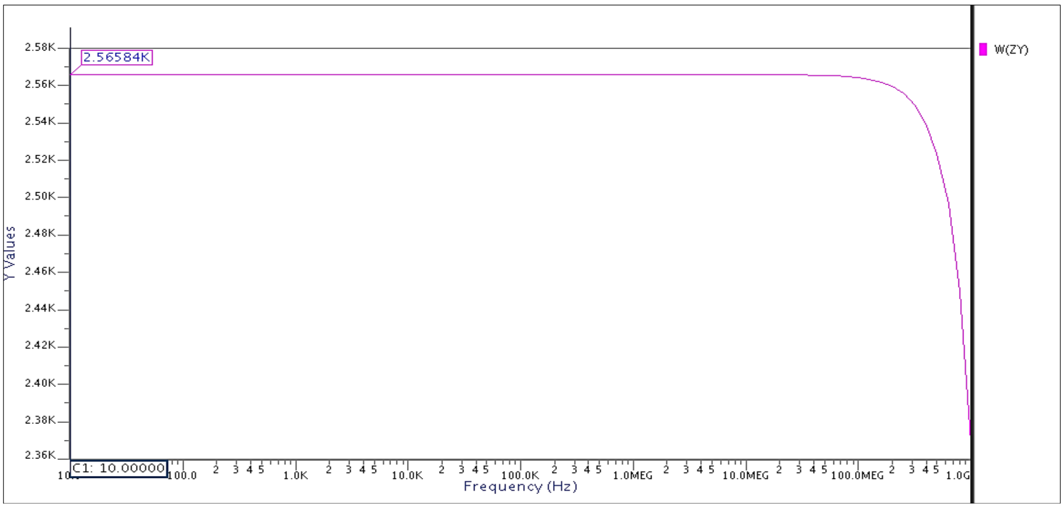

Fig. 7: *Frequency Response of the Zy*

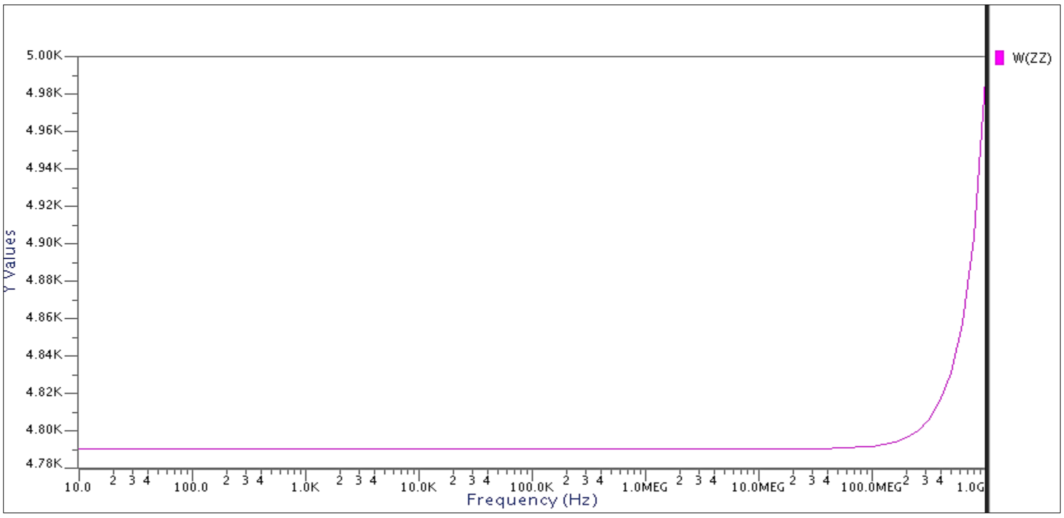

Fig. 8: *Frequency Response of the Zz.*

From the above Figure, it is shown that X port, Y port and Z port impedance respectively 1.8 K, 2.5 K and 4.79 K has been achieved.The DC transfer characteristics of Vx-Vy and Ix-Iz shown in Figure 9 and10 respectively.

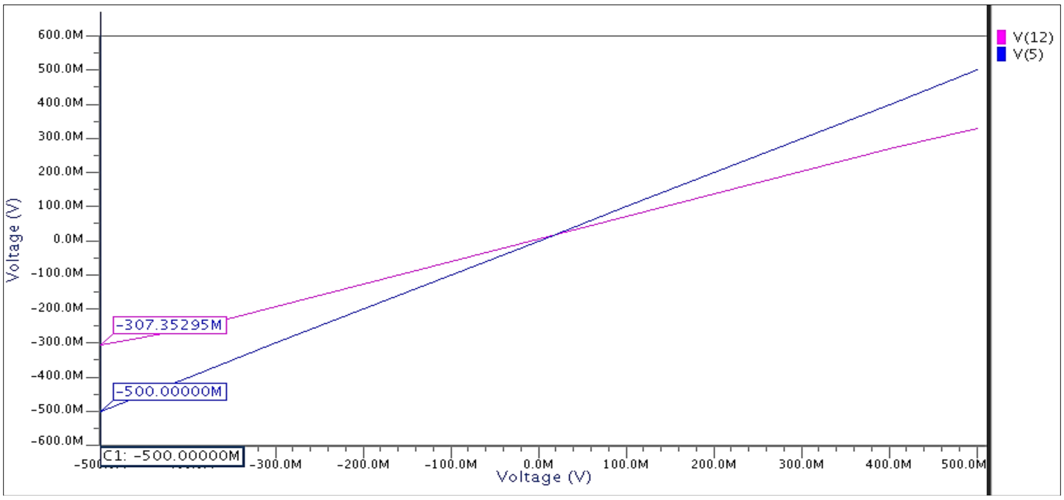

Fig. 9: *Relation Between Vx-Vy.*

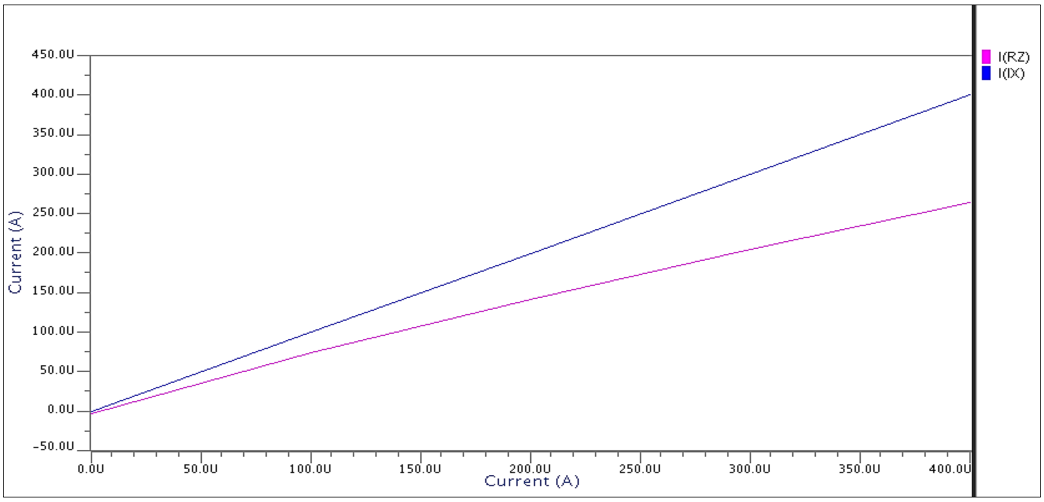

Fig. 10: *Relation Between Ix-Iz.*

From the above Figure, it is shown that voltage dynamic range and cutternt dynamic range are respectively -0.30 V to +0.30 V and -0.25mA to +0.25 mA has been achieved [9].

The AC transfer characteristics of voltage gain and current gain of cascoded CCIII is shown in Figures 11 and 12 respectively.

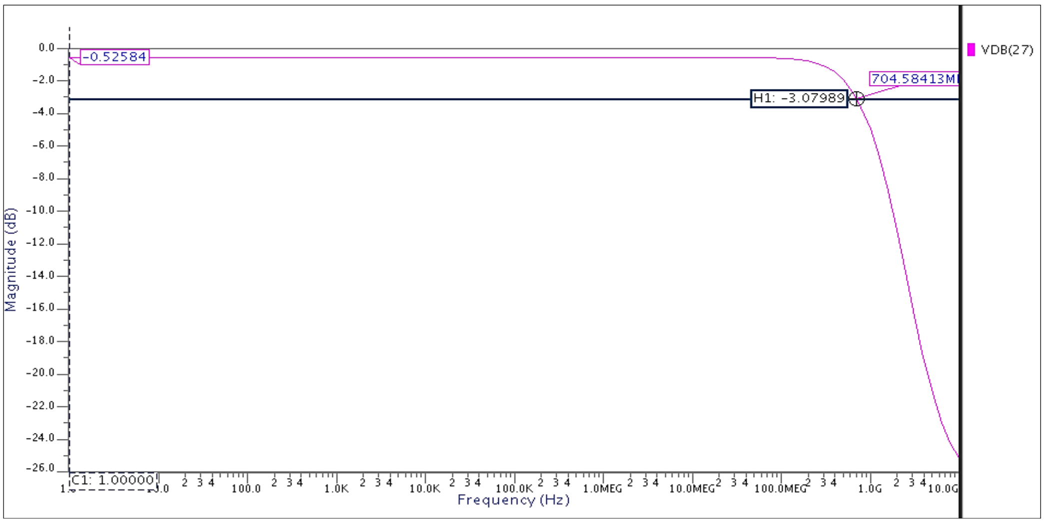

Fig. 11: *Frequency Response of the Vx/Vy.*

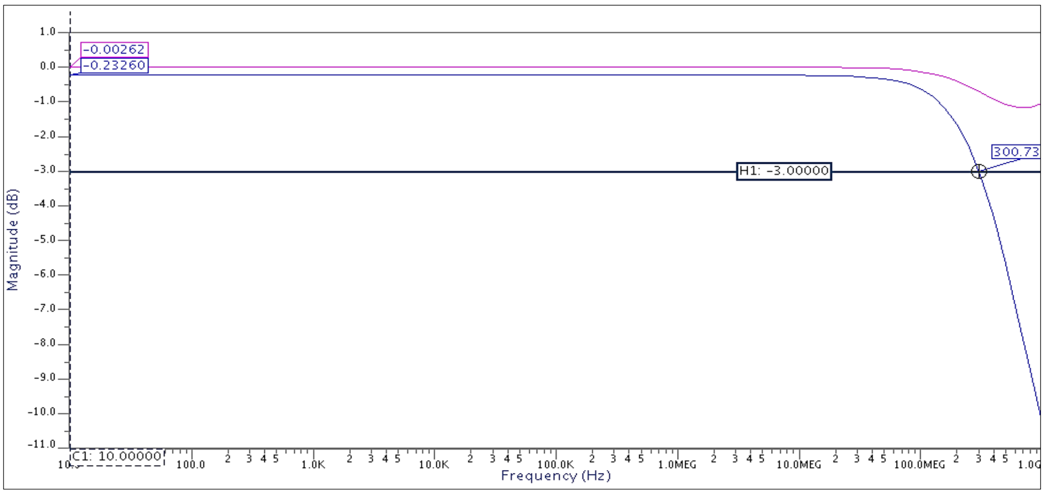

Fig. 12: *Frequency Response of the Iz+/Ix and Iz-/Ix.*

From the above Figure, it is shown that voltage bandwidth is 704 MHz and cutternt bandwidths are respectively 704 MHz and 300 MHz and > 1GHz has been achieved using DOCCIII.And X-port,Y-port and Z-port impedance is shown in Figure13, 14 and15 respectively.

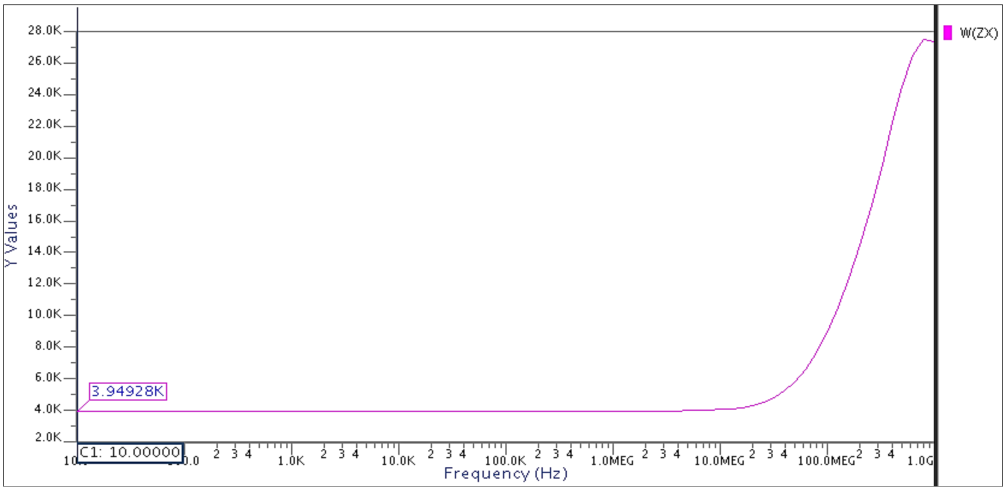

Fig. 13: *Frequency Response of the Zx.*

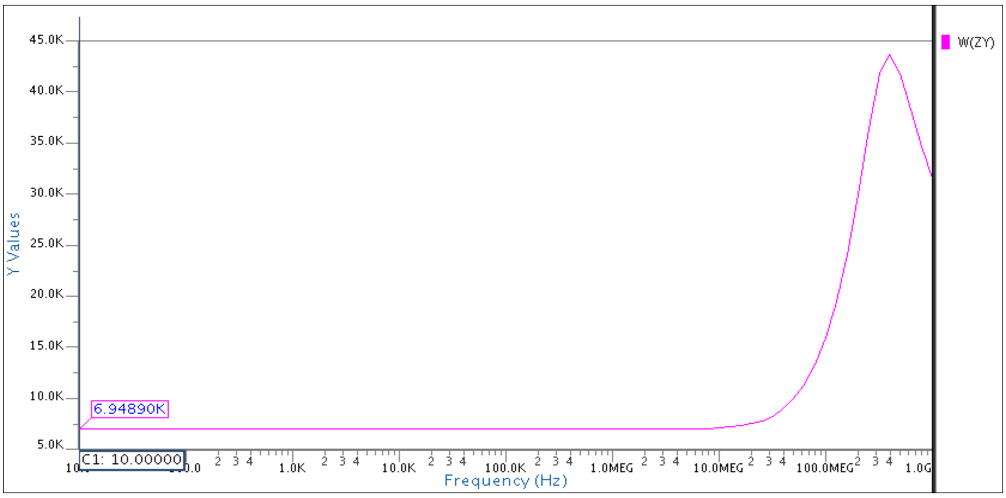

Fig. 14: *Frequency Response of the Zy.*

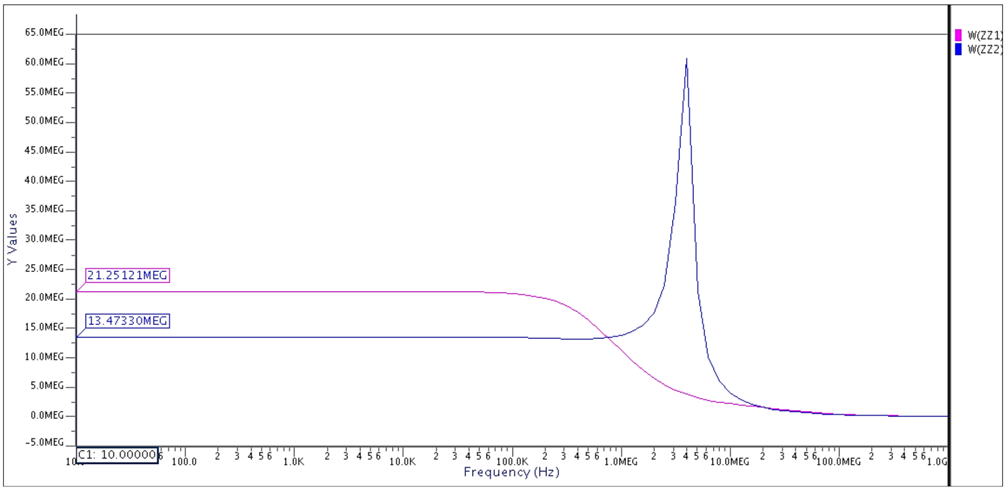

Fig. 15: *Frequency Response of the Zz+ and Zz-*

From the above Figure, it is shown that X port and Y port impedances are respectively 3.94 K, 6.94 K and Z port impedences are 21.25 MEG abd 13.47 MEG has been achieved using DOCCIII [9]. The DC transfer characteristics of Vx-Vy, Ix-Iz+ and Iz- shown in Figure 16, 17 and 18, respectively.

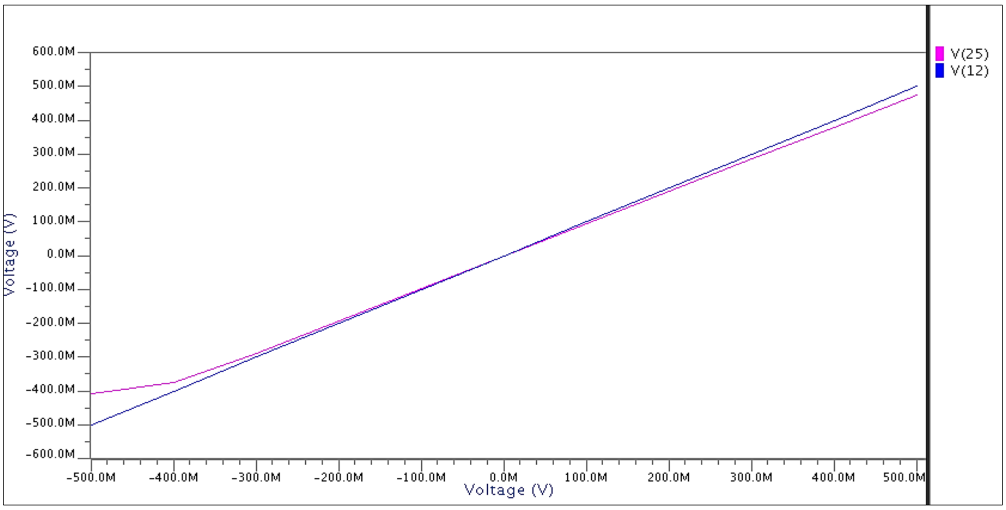

Fig. 16: *Relation between Vx-Vy.*

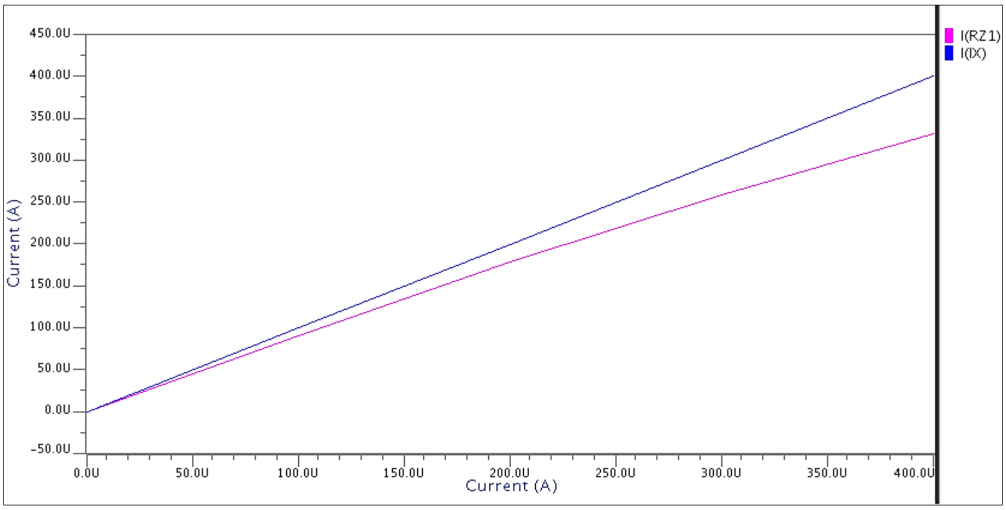

Fig.17:*Relation between Ix-Iz+.*

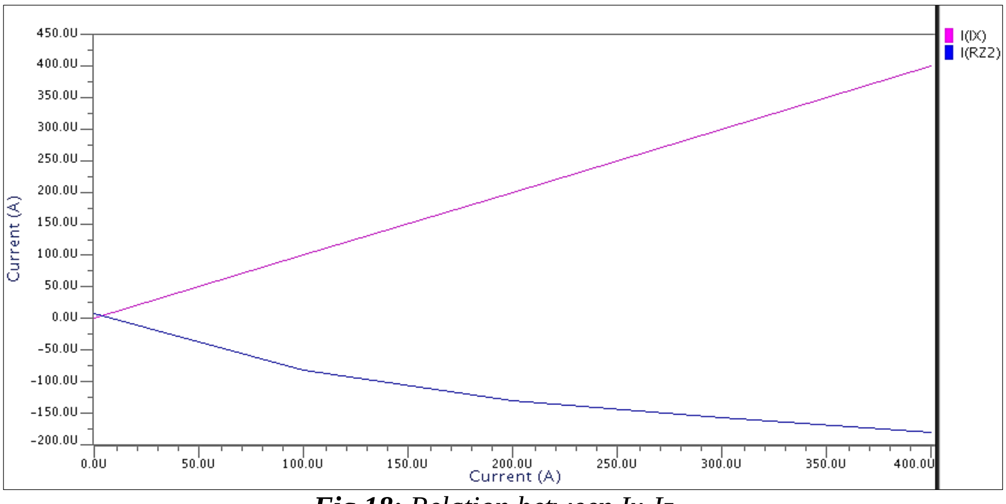

Fig.18: *Relation between Ix-Iz-.*

From the above Figure, it is shown that voltage dynamic range and cutternt dynamic range are respectively -0.41V to +0.41V and -0.35mA to +0.35 mA has been achieved using DOCCIII.

SUMMARY OF MEASUREMENT RESULTS

By comparing the conventional third generation current conveyer with the reference paper, following results are obtained.

Table I :*Summary of Conventional CCIII Simulation Results.*

PARAMETERS	CCIII[9]	CCIII
Technology (µm)	1.2	0.09
Supply Voltage (V)	2.5	0.8
Dynamic swing Vx-Vy (V)	-0.55-0.6	-0.307-0.307
Dynamic swing Ix-Iz (mA)	-0.8 - 0.8	-0.25-0.25
Vx/Vy f_{-3dB}	306 MHz	1.297 GHz
Iz+/Ix and Iz-/Ix f_{-3dB}	106 MHz	14.96 GHz
Output Resistance	18.2K	4.79K
Power dissipation	0.88 mW	7.65 µW

From the above table it is conclude that at low voltage supply power consumption is reduced and also voltage and current bandwidths are increased using trans linear loop.

By comparing the dual output cascaded current mirror third generation current conveyer with the reference paper, following results are obtained.

Table II: *Summary of Dual Output Cascoded Current Mirrors CCIII Simulation Results.*

PARAMETERS	DOCCIII[9]	DOCCIII
Technology (µm)	1.2	0.09
Supply Voltage (V)	2.5	0.8
Dynamic swing Vx-Vy (V)	-1.8-1.9	-0.41-0.41
Dynamic swing Ix-Iz (mA)	-0.65 - 0.7	-0.35 - 0.35
Vx/Vy f_{-3dB}	87MHz	0.704GHz
Iz+/Ix and Iz-/Ix f_{-3dB}	15MHz, 10.2MHz	0.300GHz, >1GHz
Output Resistance	18MEG, 17.9MEG	21.25 MEG, 13.47MEG
Power dissipation	1.37 mW	6.94 µW

From the above table it is conclude that, at low voltage supply power consumption is reduced and also voltage and current bandwidths are increased and high output impedence has been achieved because of cascaded current mirror [10].

CONCLUSION

The conventional CCIII has the main advantages of simplicity in design but it does not give high impedance. Here, conventional CCIII and cascaded current mirrors dual output CCIII are simulated in ELDO SPICE Mentor Graphics using 90 nm CMOS technology at 0.8 V supply voltage. High performance cascoded current mirrors provide high output impedance. Third generation current conveyer has the advantages of wide current and voltage bandwidths [8–10]. By the use of high performance dual output cascoded current mirrors provides high output impedance 21.25 MΩ and 13.47 MΩ instesd of in conventional CCIII output impedance 4.79 KΩ and volatage bandwidth is 704 MHz and current bandwidths are 300MHz and > 1 GHz.

In cascoded dual output third generation current conveyor power dissipation is 6.94 µW.To improve Output Impedance of dual output cascoded current mirrors are used,whichincreased output impedance of Dual output cascoded current mirrors five times greater than the conventional CCIII. The simulation results of DOCCIII confirm high performance of the circuitsinterms of linearity, voltage and current gain accuracy, power dissipation etc.

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